

MISSION 3

Part I. Problems :

Problem 1.

Use Google , deepseek ,ect... to understand the Bifurcation diagram of $D_4^-(pyramid)$.

Problem 2.

Ilya thinks that the Bifurcation diagram of $\Sigma_2(maxwell)$ is not totally correct.
Maybe we lost one more plane.

Problem 3.

Given a symmetric system :

$$\begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

Find the first Integral $I(x, y)$

Remark :

A symmetric system is a system that satisfies : P even for x and Q odd for x .

Part II. Solving :

Problem 1:

Definition : (D_4 unfolding)

The standard form of D_4 unfolding is defined as :

$$F(x, y) := \frac{1}{3}x^3 - xy^2 + \lambda_1(x^2 + y^2) + \lambda_2x + \lambda_3y$$

Where $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$.

Remark:

The bifurcation diagram of F is like a pyramid.

Problem 2:

After fixing , the bifurcation diagram of Σ_2 is :

$$A_1 = 0 \text{ or } (B_2 = 0, (A_1 - B_1)(A_1 + 3B_1) < 0)$$

Proof:

In MISSION 2 , We used the *Groebner Basis* method in *Mathematica* got the result of Σ_2 is :

$$A_1 B_2 = 0 \quad (*)$$

Actually , (*) is not totally correct , because *Groebner Basis* is doing Elimination in $\mathbb{C}$

But We need in $\mathbb{R}$.

Thus , actually the correct Σ_2 is the subset of (*).

So we will narrow the scope in two cases : $A_1 = 0$ and $B_2 = 0$.

Our idea is using Polar coordinate.

WE NEED TO INTRODUCE POLAR COORDINATE FIRST :

Using Polar coordinate :

$$\begin{cases} x = r \cos(\theta) \\ y = r \sin(\theta) \end{cases}$$

Then , the Hamiltonian fonction $H(x, y) \rightarrow H(r \cos(\theta), r \sin(\theta))$

Suppose $g(r, \theta) := (r \cos(\theta), r \sin(\theta))$

Then , $H(r \cos(\theta), r \sin(\theta)) = H(g(r, \theta)) = H \circ g(r, \theta)$

Denote $\tilde{H} := H \circ g$

Now , we have one Definition and one Theroem to introduce to you:

Definition : (*balanced point*)

Given a Hamiltonian fuction $H(x, y)$, we say that a point $(\tilde{x}, \tilde{y})$ is a *balanced point* of H if :

$$\begin{cases} H_x(\tilde{x}, \tilde{y}) = 0 \\ H_y(\tilde{x}, \tilde{y}) = 0 \end{cases}$$

Theorem :

Given a Hamiltonian function $H(x, y)$, and a Polar corrdinate as :

$$\begin{cases} x = r \cos(\theta) \\ y = r \sin(\theta) \end{cases}$$

Then :

$$(\tilde{x}, \tilde{y}) \text{ is a balanced point of } H \iff (\tilde{r}, \tilde{\theta}) \text{ is a balanced point of } \tilde{H}$$

where :

$$\begin{cases} \tilde{x} = \tilde{r} \cos(\tilde{\theta}) \\ \tilde{y} = \tilde{r} \sin(\tilde{\theta}) \end{cases}$$

PROOF OF THE THEROREM :

On one hand:

$$\mathcal{D}_{\tilde{H}(r, \theta)} = \begin{bmatrix} \frac{\partial \tilde{H}}{\partial r} & \frac{\partial \tilde{H}}{\partial \theta} \end{bmatrix}$$

On the other hand:

$$\begin{aligned} \mathcal{D}_{\tilde{H}(\tilde{r}, \tilde{\theta})} &\stackrel{\text{chain rule in inf3}}{=} \mathcal{D}_{H(g(r, \theta))} \circ \mathcal{D}_{g(r, \theta)} \\ &= \begin{bmatrix} \frac{\partial H}{\partial x}(r \cos(\theta), r \sin(\theta)) & \frac{\partial H}{\partial y}(r \cos(\theta), r \sin(\theta)) \end{bmatrix} \circ \begin{bmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{bmatrix} \end{aligned}$$

Therefore :

$$\begin{bmatrix} \frac{\partial \tilde{H}}{\partial r} & \frac{\partial \tilde{H}}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \frac{\partial H}{\partial x}(r \cos(\theta), r \sin(\theta)) & \frac{\partial H}{\partial y}(r \cos(\theta), r \sin(\theta)) \end{bmatrix} \circ \begin{bmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{bmatrix} \quad (*)$$

($\Rightarrow$) :

Suppose $(\tilde{x}, \tilde{y})$ is a balanced point of H

Then by definition :

$$\begin{cases} H_x(\tilde{x}, \tilde{y}) = 0 \\ H_y(\tilde{x}, \tilde{y}) = 0 \end{cases}$$

By (*) implies that :

$$\begin{cases} \frac{\partial \tilde{H}}{\partial r}(\tilde{r}, \tilde{\theta}) = 0 \cdot \cos(\tilde{\theta}) + 0 \cdot \sin(\tilde{\theta}) = 0 \\ \frac{\partial \tilde{H}}{\partial \theta}(\tilde{r}, \tilde{\theta}) = 0 \cdot (-\tilde{r} \sin(\tilde{\theta})) + 0 \cdot (\tilde{r} \cos(\tilde{\theta})) = 0 \end{cases}$$

Therefore , $(\tilde{r}, \tilde{\theta})$ is a balanced point of $\tilde{H}$

($\Leftarrow$) :

Suppose $(\tilde{r}, \tilde{\theta})$ is a balanced point of $\tilde{H}$

Then by definition :

$$\begin{cases} \tilde{H}_r(\tilde{r}, \tilde{\theta}) = 0 \\ \tilde{H}_\theta(\tilde{r}, \tilde{\theta}) = 0 \end{cases}$$

In (*) , since the matirx :

$$\begin{bmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{bmatrix}$$

is invertible($\det = r > 0$)

So, (*) implies that :

$$\left[\frac{\partial H}{\partial x}(r \cos(\theta), r \sin(\theta)) \quad \frac{\partial H}{\partial y}(r \cos(\theta), r \sin(\theta)) \right] = [0 \quad 0]$$

Therefore :

$$\begin{cases} H_x(\tilde{x}, \tilde{y}) = 0 \\ H_y(\tilde{x}, \tilde{y}) = 0 \end{cases}$$

Therefore , $(\tilde{x}, \tilde{y})$ is a balanced point of H

THEOREM IS DONE.

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Last time , we have the Hamiltonian function $H(x, y)$ is :

$$H(x, y) = -\frac{1}{2}(x^2 + y^2) + (B_1 - A_1)x^2y - \left(\frac{B_1}{3} + A_1\right)y^3 + B_2xy^2 - \frac{B_2}{3}x^3$$

Then :

$$\begin{cases} -\frac{1}{2}(x^2 + y^2) = -\frac{1}{2}r^2 \\ -\frac{B_2}{3} + B_2xy^2 = -\frac{B_2}{3}r^3 \cos^3(\theta) + B_2r^3 \cos(\theta) \sin^2(\theta) \stackrel{\text{Simplify}}{=} -\frac{1}{3}B_2r^3 \cos(3\theta) \\ (B_1 - A_1)x^2y - \left(\frac{B_1 + 3A_1}{3}\right)y^3 = (B_1 - A_1)r^3 \cos^2(\theta) \sin(\theta) - \left(\frac{B_1 + 3A_1}{3}\right)r^3 \sin^3(\theta) \stackrel{\text{Simplify}}{=} r^3(-A_1 \sin(\theta) + \frac{B_1}{3} \sin(3\theta)) \end{cases}$$

Therefore :

$$H(r, \theta) = -\frac{1}{2}r^2 + r^3(-A_1 \sin(\theta) + \frac{B_1}{3} \sin(3\theta) - \frac{B_2}{3} \cos(3\theta))$$

Define :

$$h(\theta) := -A_1 \sin(\theta) + \frac{B_1}{3} \sin(3\theta) - \frac{B_2}{3} \cos(3\theta)$$

That is :

$$H(r, \theta) = -\frac{1}{2}r^2 + r^3 h(\theta)$$

Now , by the definition of Σ_2 (in Polar form) :

$$\begin{cases} H_r(r_1, \theta_1) = 0 \\ H_r(r_2, \theta_2) = 0 \\ H_\theta(r_1, \theta_1) = 0 \\ H_\theta(r_2, \theta_2) = 0 \\ H(r_1, \theta_2) = H(r_2, \theta_2) \end{cases} \quad (**)$$

We have :

$$\begin{cases} H_r = -r(-1 + 3rh(\theta)) \\ H_\theta = r^3 h'(\theta) \end{cases}$$

Since $r > 0$

$$\text{Then } \begin{cases} H_r = 0 \Rightarrow r = \frac{1}{3h(\theta)} > 0 \\ H_\theta = 0 \Rightarrow h'(\theta) = 0 \\ H(r, \theta) = -\frac{1}{2}\left(\frac{1}{3h(\theta)}\right)^2 + \left(\frac{1}{3h(\theta)}\right)^3 h(\theta) \stackrel{\text{Simplify}}{=} -\frac{1}{54h^2(\theta)} \end{cases}$$

Therefore :

$$(**) \iff \exists \text{different } \theta_1, \theta_2 \text{ s.t } \begin{cases} h'(\theta_1) = h'(\theta_2) = 0 \\ h(\theta_1) = h(\theta_2) > 0 \end{cases}$$

It is amazing that Σ_2 doesn't depend on r ! it is only dependent on θ .

• **CASE I : $A_1 = 0$**

Now , our hypothesis is : $A_1 = 0$

$$\text{Then : } \begin{cases} h(\theta) = \frac{B_1}{3} \sin(3\theta) - \frac{B_2}{3} \cos(3\theta) \\ h'(\theta) = B_1 \cos(3\theta) + B_2 \sin(3\theta) \end{cases}$$

We suppose $R := \sqrt{B_1^2 + B_2^2} > 0$

(because if $=0$ it is the trivial case that $A_1, B_1, B_2 = 0$ the only solution is $(0, 0, 0).i.e(0, 0, 0) \in \Sigma_2$)

Now ,

$$\begin{cases} h(\theta) = \frac{B_1}{3} \sin(3\theta) - \frac{B_2}{3} \cos(3\theta) = \frac{R}{3} \sin(3\theta - \phi) \\ h'(\theta) = R \cos(3\theta - \phi) \end{cases}$$

$$\text{Where : } \begin{cases} \cos(\phi) = \frac{B_1}{R} \\ \sin(\phi) = \frac{B_2}{R} \end{cases}$$

$$h'(\tilde{\theta}) = 0 \stackrel{R>0}{\Rightarrow} \cos(3\tilde{\theta} - \phi) = 0 \Rightarrow 3\tilde{\theta} - \phi = \frac{\pi}{2} + k\pi \text{ where } k \in \mathbb{Z}$$

$$\Rightarrow h(\tilde{\theta}) = \frac{R}{3} \sin(3\tilde{\theta} - \phi) = \frac{R}{3} \sin\left(\frac{\pi}{2} + k\pi\right) = \frac{R}{3} (-1)^k$$

$$\Rightarrow h(\theta) = \begin{cases} \frac{R}{3} & \text{when } k \text{ is even} \\ -\frac{R}{3} & \text{when } k \text{ is odd} \end{cases}$$

But $h'(\tilde{\theta}) = 0$ has 6 solutions in $[0, 2\pi)$, 3 correspond to k even , 3 correspond to k odd

So we just need to take two of k that k even.

This tell us that $(**)$ always has solution when $A_1 = 0$.

So the plane $A_1 = 0$ is all ok !

• **CASE II : $B_2 = 0$**

$$\text{Then : } \begin{cases} h(\theta) = -A_1 \sin(\theta) + \frac{B_1}{3} \sin(3\theta) \\ h'(\theta) = -A_1 \cos(\theta) + \frac{B_1}{3} \cos(3\theta) \stackrel{\text{Simplify}}{=} \cos(\theta)(4B_1 \cos^2(\theta) - (A_1 + 3B_1)) \end{cases}$$

$$\text{So , } h'(\tilde{\theta}) = 0 \iff \cos(\tilde{\theta}) = 0 \text{ or } (4B_1 \cos^2(\tilde{\theta}) - (A_1 + 3B_1)) = 0$$

If $\cos(\tilde{\theta}) = 0$:

$$\Rightarrow \tilde{\theta} = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

But $h(\frac{\pi}{2}) = h(\frac{3\pi}{2}) > 0$ is imposible

$$\text{Because } h(\frac{\pi}{2}) = -A_1 - \frac{B_1}{3}, h(\frac{3\pi}{2}) = A_1 + \frac{B_1}{3}$$

$$\Rightarrow 0 = A_1 + \frac{B_1}{3} > 0, \text{ contradiction ! So in this subcase , no solution}$$

If $(4B_1 \cos^2(\tilde{\theta}) - (A_1 + 3B_1)) = 0$:

$$\text{We have } B_1 \neq 0 \text{ (if } B_1 = 0 \stackrel{\text{subcase}}{\Rightarrow} A_1 = 0 \Rightarrow A_1, B_1, B_2 = 0 \text{ it is the trival case)}$$

So we have $B_1 \neq 0$

$\Rightarrow$

$$\cos^2(\tilde{\theta}) = \frac{A_1 + 3B_1}{4B_1}$$

$\Rightarrow$

$$0 \leq \frac{A_1 + 3B_1}{4B_1} \leq 1$$

If $\frac{A_1 + 3B_1}{4B_1} = 0$:

$$\Rightarrow \cos^2(\tilde{\theta}) = 0 \Rightarrow \tilde{\theta} = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}. \text{ It is the case above , impossible !}$$

$$\text{So , } 0 < \frac{A_1 + 3B_1}{4B_1}$$

If $\frac{A_1 + 3B_1}{4B_1} = 1$:

$$\Rightarrow \cos^2(\tilde{\theta}) = 1 \Rightarrow \tilde{\theta} = 0 \text{ or } \pi, \text{ But } h(0) = h(\pi) = 0 \neq 0, \text{ impossible !}$$

$$\text{So , } \frac{A_1 + 3B_1}{4B_1} < 1$$

Therefore ,

$$0 < \frac{A_1 + 3B_1}{4B_1} < 1$$

Now , let $t := \cos(\tilde{\theta})$

$$\text{Then : } t^2 = \frac{A_1 + 3B_1}{4B_1}$$

So ,

$$\begin{aligned} h(\tilde{\theta}) &= -A_1 \sin(\tilde{\theta}) + \frac{B_1}{3} \sin(3\tilde{\theta}) \\ &= -A_1 \sin(\tilde{\theta}) + \frac{B_1}{3} (3 \sin(\tilde{\theta}) - 4 \sin^3(\tilde{\theta})) \\ &= -A_1 \sin(\tilde{\theta}) + \frac{B_1}{3} (\sin(\tilde{\theta})(4 \cos^2(\tilde{\theta}) - 1)) \\ &= -A_1 \sin(\tilde{\theta}) + \frac{B_1}{3} (\sin(\tilde{\theta})(4t^2 - 1)) \end{aligned}$$

$$\begin{aligned}
&= \sin(\tilde{\theta}) \left(-A_1 + \frac{B_1}{3} (4t^2 - 1) \right) \\
&= \sin(\tilde{\theta}) \left(-A_1 + \frac{B_1}{3} \left(\frac{A_1 + 2B_1}{B_1} \right) \right) \\
&= \sin(\tilde{\theta}) \left(-A_1 + \frac{A_1 + 2B_1}{3} \right) \\
&= \frac{2}{3} (B_1 - A_1) \sin(\tilde{\theta})
\end{aligned}$$

Now , consider the equation :

$$\cos^2(\tilde{\theta}) = \frac{A_1 + 3B_1}{4B_1}$$

$\tilde{\theta}$ has 4 solutions in $[0, \pi)$, 2 of them has $\sin > 0$, 2 of them has $\sin < 0$

Thus , $\begin{cases} \text{If } (B_1 - A_1) > 0 & \text{choose the first 2 of them.} \\ \text{If } (B_1 - A_1) < 0 & \text{choose the second 2 of them.} \end{cases}$

It tell us that :

$$0 < \frac{A_1 + 3B_1}{4B_1} < 1 \Rightarrow (**) \text{ always has solutions !}$$

To simplify it :

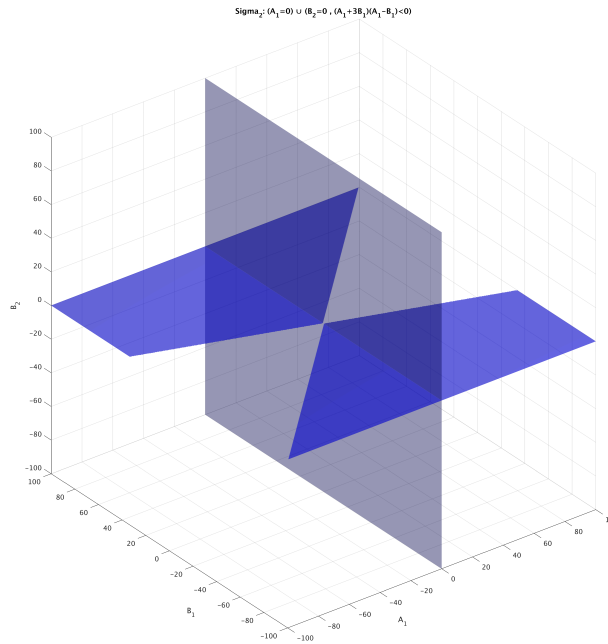
$$0 < \frac{A_1 + 3B_1}{4B_1} < 1 \iff (A_1 + 3B_1)(A_1 - B_1) < 0$$

Therefore , conclude two CASES. The bifurcation diagram of Σ_2 is :

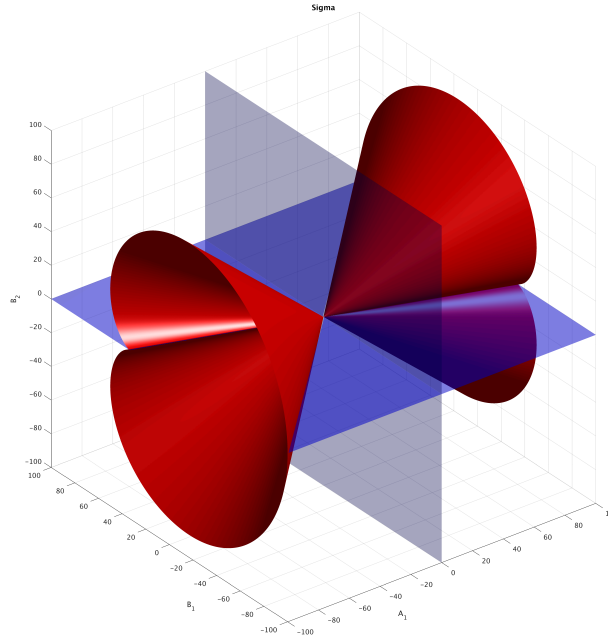
$$A_1 = 0 \text{ or } (B_2 = 0, (A_1 - B_1)(A_1 + 3B_1) < 0)$$

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The graph of Σ_2 is as follow :



The graph of Σ is as follow :



Problem 3:

We have that system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

Is symmetric.

CASE I : A is a Hamiltonian system

We know that the Hamiltonian function $H(x, y)$ is a first integral

So ,

in this case the answer is the Hamiltonian function $H(x, y) = I(x, y)$

CASE II : A is not a Hamiltonian system

Suppose our system is :

$$A : \dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

Where $A, B, C \in \mathbb{C}$

Write A into original form :

$$\begin{cases} \dot{x} = P(x, y) = -y + A_1(x^2 - y^2) - 2xyA_2 + B_1(x^2 - y^2) + 2xyB_2 + C_1(x^2 + y^2) \\ \dot{y} = Q(x, y) = x + 2xyA_1 + A_2(x^2 - y^2) - 2xyB_1 + B_2(x^2 - y^2) + C_2(x^2 + y^2) \end{cases}$$

Since A is symmetric , which means P is even and Q is odd.

Then , we have :

$$\begin{cases} A_2 = 0 \\ B_2 = 0 \\ C_2 = 0 \end{cases}$$

This tell us an important fact that :

Without losing generality , we can assume $A, B, C \in \mathbb{R}$

Then , our system A becomes :

$$A : \begin{cases} \dot{x} = P(x, y) = -y + (A + B + C)x^2 + (C - A - B)y^2 \\ \dot{y} = Q(x, y) = x + 2(A - B)xy \end{cases}$$

and since our hypothesis is A is not Hamiltonian , so :

$$2A + C \neq 0$$

Now , we are going to solve the system A

$$\text{Let } u := x^2$$

$$\Rightarrow \dot{u} = 2x\dot{x}$$

$\Rightarrow$

$$A : \begin{cases} \dot{u} = 2x(-y + (A + B + C)u + (C - A - B)y^2) \\ \dot{y} = x(1 + 2(A - B)y) \end{cases}$$

CASE 1 : If $x = 0$

Then we have the trivial solution :

$$(x, y) = (0, 0)$$

CASE 1 : If $x \neq 0$

$\Rightarrow$

$$\frac{du}{dy} = \frac{2(-y + (A + B + C)u + (C - A - B)y^2)}{1 + 2(A - B)y}$$

$\Longleftrightarrow$

$$u' + \left(\frac{-2(A + B + C)}{1 + 2(A - B)y}\right)u = \frac{2(C - A - B)y^2 - 2y}{1 + 2(A - B)y}$$

This is the one order non homogenous ODE :

$$u' + p(y)u = q(y) \quad (***)$$

with

$$\begin{cases} p(y) = \left(\frac{-2(A+B+C)}{1+2(A-B)y}\right) \\ q(y) = \frac{2(C-A-B)y^2-2y}{1+2(A-B)y} \end{cases}$$

We can solve it by using integral factor :

$$\mu(y) := e^{\int p(y)dy} = e^{\int \left(\frac{-2(A+B+C)}{1+2(A-B)y}\right)dy}$$

After solving (***) , We get the first integral are :

$$I(x, y) = \begin{cases} e^{-2Ky}(x^2 + \frac{E}{K}y^2 + (\frac{E}{K^2} - \frac{1}{K})y + \frac{E}{2K^3} - \frac{1}{2K^2}) & \text{When } D = 0 \\ (1 + 2Dy)^{-\frac{K}{D}}(x^2 - \frac{E(1+2Dy)^2}{4D^2(2D-K)} + \frac{(E+D)(1+2Dy)}{2D^2(D-K)} + \frac{E+2D}{4D^2K}) & \text{When } D \neq 0, K \neq 0, K \neq D, K \neq 2D \\ x^2 - \frac{1}{4D^3}(\frac{E}{2}(1 + 2Dy)^2 - 2(E + D)(1 + 2Dy) + (E + 2D) \ln |1 + 2Dy|) & \text{When } D \neq 0, K = 0 \\ \frac{x^2}{1+2Dy} - \frac{1}{4D^3}(E(1 + 2Dy) - 2(E + D) \ln |1 + 2Dy| - \frac{E+2D}{1+2Dy}) & \text{When } D \neq 0, K = D \\ \frac{x^2}{(1+2Dy)^2} - \frac{1}{4D^3}(E \ln |1 + 2Dy| + \frac{2(E+D)}{1+2Dy} - \frac{E+2D}{2(1+2Dy)^2}) & \text{When } D \neq 0, K = 2D \end{cases}$$

Where :

$$\begin{cases} D := A - B \\ K := A + B + C \\ E := C - A - B \end{cases}$$

Done.